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Rahimah Bay Area Biodiversity Baseline Survey and Conservation Strategy for a Sustainable & Circular Tomorrow

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ABSTRACT

The objective of this environmental initiative is to develop a comprehensive management and conservation plan for the Ras Tanura Producing Department (RTPD), Rahimah Bay Biodiversity Protected Area in Saudi Arabia, a unique and diverse habitat comprising terrestrial, coastal, and marine ecosystems. The aim is to preserve the area's ecological integrity and promote biodiversity conservation and circular economy, aligning with the Saudi Aramco's strategic objectives and the UN Sustainable Development Goals (SDGs). The project demonstrates a modern, sustainable, and multi-disciplinary approach to target biodiversity threats and transform them into opportunities to improve the Rahimah Bay Biodiversity Protected Area's ecosystem.

The analytical approach consists of the following methodology:

- Conduct a thorough survey to establish a baseline understanding of the area's biodiversity, including literature review, habitat mapping, general terrestrial surveys, marine survey, E-DNA sampling, bird surveys, breeding bird surveys, non-breeding bird surveys, wetland bird surveys, and migratory bird surveys.
- Evaluate the area's ecological characteristics, identify threats, and assess ecosystem health using remote sensing technology, ground truthing exercises, and species inventories.
- Develop targeted conservation strategies and management plans based on the survey outcomes, including the identification of conservation priorities and the development of a Biodiversity Action Plan.
- Implement initiatives to enhance biodiversity based on extensive biodiversity baseline survey, promote environmental awareness, and align with the company's strategic objectives, such as mangrove plantation, mangrove nursery construction, beach cleanup initiatives and coral reef deployment.

The conservation and management of the Rahimah Bay Biodiversity Protected Area has several positive environmental impacts. First, it helps preserve the area's unique biodiversity, including various animals, such as birds, turtles, and foxes. Second, it promotes environmental sustainability and reduces the environmental impact of human activities on the natural environment. In addition, it supports the company's commitment

to environmental stewardship and sets a precedent for similar conservation efforts in the region. Finally, it contributes to the preservation of the area's ecological integrity and promotes the well-being of local communities.

The implementation of this unique innovative approach at the Rahimah Bay Biodiversity Protected Area demonstrates a lead-by-example role in environmental aspects. The corporate commitment as a global oil and gas producer toward the environment was clearly illustrated in this initiative. Building a circular environmental culture that contributes to combating climate change and sustaining the environment falls under the company's corporate strategies. The development of a comprehensive management and conservation plan is not a new concept, but the use of a multi-disciplinary approach and innovative methods, such as remote sensing technology, E-DNA sampling and Normalized Difference Vegetation Index (NDVI), is a sustainable and effective solution. The approach can also be proposed for other conservation areas, where biodiversity preservation is a priority. This environmental initiative is a win-win solution that can benefit both the environment and human development.

INTRODUCTION

This paper presents an innovative proposal to develop a comprehensive management and conservation plan for the Rahimah Bay Biodiversity Protected Area in Saudi Arabia, a unique and diverse habitat comprising terrestrial, coastal, and marine ecosystems. This proposal propagates in the same direction of Saudi Aramco's strategy of environmental stewardship, biodiversity conservation, and sustainable development. Moreover, this innovation demonstrates a modern and sustainable approach to target biodiversity threats in multi-dimensional aspects and transform these threats into opportunities to improve the Rahimah Bay Biodiversity Protected Area's ecosystem.

The Rahimah Bay Biodiversity Protected Area is a vital habitat for various species, including birds, turtles, and foxes, and serves as a migratory path for several birds species. The area's ecological integrity can be safeguarded and enhanced through concerted conservation efforts, innovative solutions, and responsible management of human activities, such as coastal development and pollution, to maintain the delicate balance of the ecosystem and promote biodiversity. Consequently, there is a need to develop a comprehensive management and conservation plan to preserve the area's biodiversity and promote environmental sustainability. The plan will undergo a thorough survey to establish a baseline understanding of the area's biodiversity, including literature review, habitat mapping, general terrestrial surveys, marine survey, E-DNA sampling, bird surveys, breeding bird surveys, non-breeding bird surveys, wetland bird surveys, and migratory bird surveys.

The Rahimah Bay Biodiversity Protected Area presents a unique opportunity for collaborative conservation and management efforts, bringing together stakeholders to balance human needs with environmental protection. The numerous benefits of conservation are clear, including the enhancement of ecosystem services, improvement of water quality, and creation of thriving habitats for marine life. By protecting this vital area, we can also reduce the risk of coastal erosion and flooding, while promoting environmental awareness and education among local communities and visitors alike.

Until recently, biodiversity conservation was perceived as a secondary priority, however, industrial perception begun shifting towards recognizing the importance of environmental stewardship and biodiversity conservation. Saudi Aramco has been a pioneer in recognizing the importance of environmental stewardship and biodiversity conservation, with a long history of commitment to protecting the natural world. For decades, the company has been at the forefront of efforts to preserve and promote biodiversity, and its dedication to this cause has only grown stronger over time. The development of a comprehensive management and conservation plan for the Rahimah Bay Biodiversity Protected Area is a testament to Aramco's enduring commitment to environmental sustainability and its aspiration to make a positive impact on the environment.

As a leader in the industry, Aramco has consistently demonstrated its willingness to innovate and adapt in order to stay ahead of the curve in biodiversity conservation. The company's use of cutting-edge technologies, such as remote sensing and Normalized Difference Vegetation Index (NDVI) & E-DNA sampling, which are some examples of Ras Tanura Producing Department (RTPD's) proactive approach to promoting environmental sustainability and protecting the natural habitats of the Rahimah Bay area. By leveraging these innovative methods, Aramco aims to not only conserve biodiversity but also to enhance the overall health and resilience of the ecosystem.

This paper will delve into the details of Saudi Aramco, Ras Tanura Producing Department (RTPD's) comprehensive management and conservation plan for the Rahimah Bay Biodiversity Protected Area, detailed methodology of baseline survey, highlighting the company's vision, strategies, and initiatives aimed at promoting biodiversity conservation and environmental sustainability. Through its tireless efforts, Aramco is setting a new standard for environmental responsibility and stewardship, and its work in the Rahimah Bay area serves as a model for others to follow.

METHODOLOGY

Recognizing the importance of preserving precious natural resource, Ras Tanura Producing Department (RTPD), Saudi Aramco has taken proactive approach to protecting the Rahimah Bay Biodiversity Protected Area. By developing a comprehensive management and conservation plan, the company aims to not only safeguard the area's biodiversity but also promote environmental sustainability and ensure the long-term health of the ecosystem. This forward-thinking initiative will help to mitigate the impacts of human activities, such as coastal development and pollution, and create a balanced and harmonious relationship between human needs and environmental protection.

The Rahimah Bay Biodiversity Protected Area in Saudi Arabia is a vibrant and thriving ecosystem, boasting an incredible array of terrestrial, coastal, and marine habitats that support a stunning diversity of flora and fauna. This unique and fascinating region is home to a wide range of species, including birds, turtles, and foxes, and serves as a critical migratory pathway for numerous other species. As a result, the area plays a vital role in maintaining the health and resilience of the entire ecosystem.

The conservation and management of the Rahimah Bay Biodiversity Protected Area involves a thorough survey to establish a baseline understanding of the area's biodiversity, including literature review, habitat mapping, general terrestrial surveys, marine survey, E-DNA sampling, bird surveys, breeding bird surveys, non-breeding bird surveys, wetland bird surveys, and migratory bird surveys. The survey will be conducted using remote sensing technology, ground truthing exercises, and species inventories to evaluate habitat conditions, identify threats, and assess ecosystem health.

The methodology for this study will involve a combination of literature review, habitat mapping, general terrestrial surveys, marine surveys, and bird surveys as detailed below.

Literature Review

A comprehensive literature review includes collecting information on the ecology of the biodiversity protected area comprising flora, fauna, mangroves, intertidal and coastal ecosystems including seagrass beds, tidal mudflats and saltmarshes. Review includes marine and oceanographic studies of relevance to the coast of Gulf where the biodiversity protected area is located.

The literature review includes fish, crustaceans (e.g. shrimp), and bird species. Historical data on the related organisms shall be consolidated to compare and infer the results obtained in the present study. A key output of the literature review is a threat identification assessment.

Habitat Mapping

During the Habitat mapping all terrestrial, coastal and marine habitats within the boundary of the biodiversity protected area are mapped with the aim of establishing a baseline against which changes in the

extent and condition (i.e. health) of vegetation can be tracked. As a minimum, habitat mapping employs remote sensing satellite data and the Normalized Difference Vegetation Index (NDVI) supplemented by on-ground surveys.

The survey enables the development of habitat maps that provide information on the distribution and extent of natural biological communities. Habitat site detail are acquired from aerial and satellite photographs of the project site. Other site-specific data is obtained from data collection by the team responsible for mapping. This spatially explicit approach also applies to the fauna surveys in order that the distribution of species is understood, in order to inform the Biodiversity detailed Action Plan. Threats to habitat and species are also mapped and assessed for severity and likely impact on ecologically sensitive receptors

General Terrestrial Surveys

Terrestrial ecology surveys are conducted to assess the area vegetation cover, flora, and fauna in the Biodiversity Protected Area. Furthermore, the survey includes:

- Vegetation surveys to assess percentage vegetation cover
- Flora surveys to identify plant species present
- Diurnal surveys to identify vertebrate species
- Nocturnal spotlighting for vertebrates
- Remote camera trapping for vertebrates
- Ethical walk-in traps for small mammals and reptiles
- Ultrasonic surveys for micro-bats

Marine Survey

Marine survey is conducted to map marine biotypes within the biodiversity protected area. The survey utilizes a combination of survey transects using drop-down cameras, replicate quadrat sampling, and drop-down camera surveys to map benthic habitats. The objective is to develop a detailed map of marine habitats with a key focus on the extent and condition of seagrass beds. All incidental species records encountered during these surveys area also recorded.

Bird Surveys

Bird surveys are conducted during both migration seasons, winter, and summer to evaluate over-wintering populations and breeding activity. Survey includes:

- Breeding bird surveys to identify breeding bird species and their reliance on the site
- Non-breeding bird surveys to identify non-breeding bird species and their potential reliance on the site for overwintering
- Wetland bird surveys to identify wetland bird species and their reliance on the site
- Migratory bird surveys to identify migratory bird species and their reliance on the site

Survey Methods

The following survey methods are used:

1. Remote Sensing: This method involves using satellite or aerial imagery to collect data about the environment. Remote sensing can be used to track changes in vegetation extent and condition over time. NDVI (Normalized Difference Vegetation Index) is a specific type of remote sensing data that

measures the health and density of vegetation. By analyzing NDVI data, researchers can identify areas of vegetation growth or decline, which can inform conservation efforts.

2. **On-Ground Surveys:** These surveys involve collecting data directly from the field by visiting the site and observing the environment, flora, and fauna. On-ground surveys can provide detailed, site-specific information that can be used to validate remote sensing data or to gather information that cannot be collected through remote sensing.
3. **Transect Surveys:** A transect survey involves walking or traveling along a predetermined path (transect) to collect data on vegetation cover, flora, and fauna. This method allows researchers to systematically sample the environment and gather data on the distribution and abundance of different species.
4. **Quadrat Sampling:** Quadrat sampling involves dividing an area into smaller, equal-sized plots (quadrats) and collecting data on the species present within each quadrat. This method is often used to assess benthic habitats, such as coral reefs or sea grass beds, where species are closely associated with specific substrates or microhabitats.
5. **Drop-Down Camera Surveys:** This method involves using a camera attached to a rope or cable to collect images of the seafloor or other underwater environments. Drop-down camera surveys can be used to map marine biotypes, such as coral reefs or kelp forests, and to gather data on the species present in these areas.
6. **Vantage Point Surveys:** Vantage point surveys involve observing the environment from a fixed location, such as a tower or a hilltop, to gather data on migratory bird species and their reliance on the site. This method can provide information on the importance of the site as a stopover or breeding ground for different bird species.
7. **Ultrasonic Surveys:** Ultrasonic surveys use specialized equipment to detect the high-frequency sounds made by micro-bats. By analyzing these sounds, researchers can identify the presence and abundance of different bat species, which can inform conservation efforts.
8. **Ethical Walk-In Traps:** Ethical walk-in traps are a type of trap that allows small mammals and reptiles to enter voluntarily and then be released unharmed. This method is used to capture and study these animals without causing them harm or stress.
9. **Remote Camera Trapping:** Remote camera trapping involves using cameras with motion sensors or other triggers to capture images of vertebrates, such as mammals or birds, in their natural habitat. This method can provide valuable information on the presence, abundance, and behavior of different species, and can be used to monitor population trends over time.

These methods can be used individually or in combination to gather a comprehensive understanding of the environment and the species that inhabit it. By using a range of methods, researchers can gather detailed, accurate data that can inform conservation efforts and management decisions.

Data Analysis

Data analysis is conducted using international best practice norms and survey methodologies specified. All species data is recorded and mapped, and that includes information on species richness, abundance, and location.

Biodiversity Action Plan (BAP)

The biodiversity action plan is developed based on the results of the surveys and it includes the following components:

- **Threat Identification:** A threat identification assessment is conducted as part of BAP to identify potential threats to the biodiversity of the Biodiversity Protected Area.

- **Habitat Conservation:** Habitat conservation measures are included in the BAP for implementation to protect and conserve the habitats of the Biodiversity Protected Area.
- **Species Conservation:** Species conservation measures are included in the BAP for implementation to protect and conserve the species of the Biodiversity Protected Area.
- **Monitoring and Evaluation:** Monitoring and evaluation is also part of BAP to assess the effectiveness of the biodiversity action plan and to identify areas for improvement.
- **Biodiversity Actions:** Schedule of actions with priorities are included in the action plan for better tracking and implementation.
- **Community Engagement:** Community engagement is also included in BAP to raise awareness about the importance of biodiversity conservation and to involve the local community and stakeholders in the conservation efforts.

CONCLUSION

This environmental initiative employs a comprehensive management and conservation plan for the Ras Tanura Producing Department (RTPD) Saudi Aramco, Rahimah Bay Biodiversity Protected Area in Saudi Arabia. The aim of this initiative is to reduce the negative environmental impact of human activities and establish sustainable and biodiverse ecosystems, improve circularity and to combat climate change and achieve net zero emissions as part of Saudi Aramco's mission and Saudi Arabia's 2030 vision. The results of the baseline surveys are fundamental in the development of a biodiversity action plan that includes threat identification, habitat conservation, species conservation, monitoring and evaluation, and community engagement. The plan should also include measures to mitigate the impacts of human activities, such as coastal development and pollution, and promote sustainable livelihoods for local communities. The use of innovative methods, such as remote sensing technology, E-DNA sampling and NDVI, will be crucial in monitoring and evaluating the effectiveness of the conservation plan. Additionally, the involvement of local communities and stakeholders will be essential in ensuring the success of the plan and promoting environmental awareness and education.

The following conclusions can be drawn from the information presented throughout this paper:

- The Rahimah Bay Biodiversity Protected Area is a unique and diverse habitat comprising terrestrial, coastal, and marine ecosystems, requiring a comprehensive management and conservation plan to preserve its ecological integrity.
- The traditional conservation and management methods are not sufficient to address the environmental challenges faced by the Biodiversity Protected Area, and a new approach is needed to promote sustainability and biodiversity.
- The mangrove forests, can provide a positive net return on investment and help to reduce the negative environmental impact and promote circular economy.
- Mangrove forests can effectively reduce the carbon footprint and encourage ecological and biological diversification of local marine life by introducing a valuable food source, and can also provide more security to the area's structure against erosion due to tidal forces of the surrounding sea.
- The implementation of this comprehensive management and conservation plan can be proposed to external entities to shift the perception of conservation as a secondary priority into a beneficial and valuable resource with the potential to provide a positive net return on investment.
- Lastly, this environmental initiative demonstrates Saudi Aramco's commitment to environmental stewardship, biodiversity conservation, and sustainable development, and can serve as a model for similar conservation efforts in the region and around the world.

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